

Zhuoyang Zou

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Research Interests

- **Agentic AI for Science and Healthcare.** Building multi-agent LLM systems that automate real-world scientific workflows — AI-assisted peer review, clinical decision support, and disease prediction.
- **LLM Reasoning and Agent Design.** Optimizing agent architectures for complex reasoning: adaptive planning, multi-agent debate, tool use, and long-horizon task execution.

Education

Pennsylvania State University

Ph.D. in Computer Science, Advisor: Prof. Wenpeng Yin

Sep. 2024 – Present

University Park, PA

Northwestern University

M.S. in Artificial Intelligence

Sep. 2022 – Dec. 2023

Evanston, IL

The Chinese University of Hong Kong

B.S. in Computer Science, Advisor: Prof. Qi Dou

Sep. 2017 – May 2022

Hong Kong

Work Experience

Pennsylvania State University

Researcher, Agentic AI & Multi-Agent LLM Systems

Sep. 2024 – Present

University Park, PA

- **DIAGPaper** [*Python, GPT-4o, Claude, Llama, LangChain*]: Multi-agent LLM framework for automated scientific peer review. Customizer dynamically generates paper-specific criteria; Rebuttal module runs adversarial author-reviewer debate filtering 40–60% invalid critiques; Prioritizer ranks weaknesses by severity from 15,343 real reviews. **F1 51.89 on AAAR, +9.4% over GPT-4o.**
- **AAAR-1.0** [*Python, NLP pipeline, benchmark design*]: Large-scale benchmark for AI-assisted research. **ICML 2025, Workshop Best Paper Award.**
- **M2-Verify** [*VLMs, multimodal reasoning*]: Multidomain benchmark for multimodal claim consistency across text and image.
- **Bridging the Know-Act Gap** [*LLM agents, autoregressive reasoning*]: Framework closing the gap between LLM knowledge and executable action in agentic settings.

High Fashion Group

AI Specialist

May 2024 – Aug. 2024

Hong Kong

- **Fabric Recognition System** [*PyTorch, Siamese Network, ResNet*]: Few-shot image similarity model for fabric pattern matching and quality control.
- **Semantic Fabric Search** [*LLM, RAG, vector embeddings*]: Text-to-fabric retrieval using LLM query encoding and dense retrieval over product database.
- **Trend Forecasting RAG** [*RAG, LLM, structured data*]: Multi-source pipeline for fashion trend analysis and cost-performance reporting.

Northwestern University Medical School

Researcher, Medical AI

Jul. 2023 – May 2024

Chicago, IL

- **iTox** [*PyTorch, CNN-RNN, Grad-CAM*]: Pneumonitis risk prediction from pre-treatment CT scans. **AUC 0.84.** Grad-CAM localizes predictive lung regions; iTox dose metric reduces pneumonitis risk by **20%**.

Karmo International Company Ltd.

Artificial Intelligence Developer

Jun. 2022 – May 2023

Hong Kong

- **Construction Safety Detection** [*YOLOv5/v8, OpenCV*]: Real-time safety gear detection deployed at active construction sites.
- **Autonomous Vehicle Detection** [*TensorFlow, object detection*]: Multi-class real-time detection deployed at Hong Kong Science Park.
- **Drainage Prediction System** [*Django, Random Forest, scikit-learn*]: Web app and predictive model for Hong Kong Government Drainage Services.

The Chinese University of Hong Kong

Researcher, VR & Surgical AI

Jan. 2022 – Aug. 2022

Hong Kong SAR

- **VR Surgical Training System** [*Unity C#, stereoscopic rendering*]: Simulated Da Vinci surgical procedures with stereoscopic vision for trainee surgeons.
- **Surgical Instrument Segmentation** [*U-Net, PyTorch*]: Segmentation and activity recognition on surgical video; eye-tracking for adaptive training.

Publications

- **Z. Zou**, A. Ansari, D.C. Zhang, D. Lee, W. Yin. **DIAGPaper: Diagnosing Valid and Specific Weaknesses in Scientific Papers via Multi-Agent Reasoning**. *arXiv:2601.07611*, 2026. [Link]
- JJ Ahn, R. Kamoi, B. Atil, R. Lou, WW Kang, H. Park, SSS Das, **Z. Zou**, et al. **Bridging the Know-Act Gap via Task-Level Autoregressive Reasoning**. *arXiv:2603.22619*, 2026. [Link]
- A. Ansari, D.C. Zhang, **Z. Zou**, W. Yin, D. Lee. **M2-Verify: A Large-Scale Multidomain Benchmark for Checking Multimodal Claim Consistency**. *arXiv:2604.01306*, 2026. [Link]
- B. Belland, W. Yin, **Z. Zou**, J. Mussoline, C. Kim. **AI-Enabled Automated Scaffolding for Undergraduate Students' Learning to Debug and Reason**. *AI4HE @ CSEDU 2026*.
- R. Lou, H. Xu, **Z. Zou**, et al. **AAAR-1.0: Assessing AI's Potential to Assist Research**. *ICML 2025. Workshop Best Paper Award*. [Link]
- Y. Zhang, W. Zheng, **Z. Zou**, et al. **HRScene: How Far Are VLMs from Effective High-Resolution Image Understanding?** *ICCV 2025*. [Link]
- **Z. Zou**, P.T. Teo, A. Yalamanchili, M. Abazeed. **Optimizing Deep Learning Models: CNN-RNN Augmentation with Grad-CAM Analysis for Predicting Pneumonitis**. *Int. J. Radiation Oncology, Biology, Physics*, 2024. [Link]

Technical Skills

- **LLMs & Agents:** GPT-4o, Claude (Sonnet/Opus), Llama 3, Qwen, Gemini, LangChain, LangGraph, AutoGen, RAG, tool use, chain-of-thought, prompt engineering, multi-agent orchestration
- **AI / ML:** PyTorch, TensorFlow, HuggingFace Transformers, CNN, RNN, YOLO, U-Net, Siamese Networks, Grad-CAM, scikit-learn
- **Dev & Tooling:** Python, Claude Code, Cursor, GitHub Copilot, Docker, Git, Linux, FastAPI, Django, Unity C#
- **Data & Infra:** FAISS, ChromaDB, SQL, AWS, Weights & Biases, Jupyter